

# Sathwika Meenuga

## Data Analyst

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### Professional Summary:

- Results-driven Data Analyst with 4 years of extensive experience in the healthcare and financial sectors.
- Skilled in integrating real-time fraud detection algorithms into operational systems via AWS Lambda, enhancing organizational security through effective alert systems.
- Proficient in extracting insights from large datasets using SQL, Teradata, Hadoop/HIVE/Sqoop and Apache Spark, enabling data-driven decision-making across diverse business contexts.
- Strong understanding of Agile/Scrum methodologies, adept at translating complex business requirements into technical solutions, and optimizing the performance of TensorFlow models.
- Committed to adding value to teams through innovative data solutions while proactively addressing potential disruptions in data integrity and accessibility.
- Led compliance initiatives for loan reviews, ensuring alignment with lender requirements while managing data deliverables and high-level data modelling.
- Streamlined data cleansing processes and reporting through advanced Excel techniques and SQL, producing complex reports to audit healthcare assets effectively.
- Leveraged Business Intelligence tools like Power BI to create dynamic dashboards and ad hoc reports, facilitating data-driven solutions to business challenges.
- Experienced in data pre-processing and statistical analysis with Python, utilizing libraries such as Pandas and NumPy, and conducting A/B testing to refine marketing strategies and improve user engagement.
- Expertise in developing and maintaining comprehensive data dictionaries and definitions utilizing Collibra, alongside proficiency in statistical analysis and data visualization through Python and R, ensuring clarity and accessibility of data.
- Proven ability to design and refine healthcare data models using SQL, significantly enhancing analytical capabilities and improving forecasting accuracy in healthcare settings.
- Extensive knowledge of HIPAA compliance and various medical plans, including HMO, PPO, POS, EPO, and Medicare/Medicaid, ensuring adherence to regulatory standards in data handling.
- Enhanced organizational decision-making processes by automating data access and reporting, leading to increased operational efficiency and faster business development cycles.
- Developed predictive models that effectively addressed customer retention challenges, contributing to revenue growth through improved customer engagement strategies.

### Technical Skills:

|                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|-------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Methodologies:</b>         | SDLC, Agile (Scrum), Waterfall.                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>Programming Languages:</b> | Python, SQL, R.                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>Packages:</b>              | Scikit-Learn, ggplot2, Pandas, NumPy, Matplotlib, SciPy, Seaborn.                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Visualization Tools:</b>   | Tableau, Power BI, Advanced Excel (Pivot Tables, VLOOKUP, VBA, Macros)                                                                                                                                                                                                                                                                                                                                                                            |
| <b>IDEs:</b>                  | Visual Studio Code, PyCharm, Jupyter Notebook.                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>Database Management:</b>   | MySQL, MongoDB, PostgreSQL, Oracle.                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Cloud Platforms:</b>       | AWS Snowflake, AWS Redshift, AWS Databricks, AWS S3, AWS Lambda.                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Other Technologies:</b>    | SSIS, SSRS, SAS, Machine Learning Algorithms, ETL, Probability distributions, Confidence Intervals, ANOVA, Hypothesis Testing, Regression Analysis, Linear Algebra, Advance Analytics, Data Mining, Data Visualization, Data warehousing, Data transformation, Data Storytelling, Association rules, Clustering, Classification, Regression, A/B Testing, Forecasting & Modelling, Data Cleaning, Data Wrangling, MS Office Suite, Alteryx, Jira. |

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|---------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Soft Skills:</b>       | Time management, Leadership, Management, Problem-solving, Negotiation, Decision-Making, Documentation and Presentation, Verbal communication. |
| <b>Version Control:</b>   | Git, GitHub, SVN.                                                                                                                             |
| <b>Healthcare</b>         | Epic Systems, HL7, FHIR, Medicare/Medicaid data analysis, EHR.                                                                                |
| <b>Operating Systems:</b> | Windows, Linux, Mac iOS.                                                                                                                      |

### **Professional Experience:**

#### **Cigna Healthcare – CT (Remote) Data Analyst**

**Aug 2023 – Current**

- Utilized SQL for data extraction from healthcare databases containing patient records, demographics, and medical history.
- Leveraged machine learning tools like Scikit-learn and TensorFlow to create predictive models for inventory management, reducing stockouts by 15%.
- Conducted exploratory data analysis (EDA) to identify trends, patterns, and correlations in population health data, using Python libraries such as Pandas and NumPy.
- Collaborated with business stakeholders to design, develop, and optimize data-driven solutions for Medicare and Medicaid clients.
- Developed and implemented machine learning models using Scikit-learn and Keras to predict patient readmission risks, improving care planning and reducing readmission rates by 12%.
- Utilized AWS, Snowflake, and Redshift to store and analyze large-scale healthcare data, ensuring fast access to key insights and reducing query processing time by 30%.
- Developed and maintained automated ETL pipelines using SQL and Alteryx, reducing manual data processing time by 40%.
- Developed predictive models using scikit-learn in Python to forecast healthcare resource utilization, including hospital admissions, emergency room visits, and specialist consultations.
- Conducted data cleansing and transformation processes, ensuring accurate data reporting for HEDIS measures and compliance with CMS guidelines.
- Worked with Epic and HL7 data standards to integrate and process clinical data for improved reporting accuracy and completeness.
- Analyzed Medicare and Medicaid claims data to identify trends and opportunities for cost optimization and healthcare improvements.
- Designed a PyTorch model to analyze medical imaging data, achieving an accuracy rate of 95% in early-stage disease detection, facilitating timely intervention.
- Applied advanced Excel functions, such as VLOOKUP and PivotTables, for comprehensive data analysis, resulting in a 25% improvement in reporting accuracy.
- Ensured compliance with data privacy laws (HIPAA) by implementing robust data security practices, maintaining 100% compliance during internal audits.
- Understanding business requirements and providing workable technical solutions through Agile/Scrum steps after testing, user acceptance, and optimized TensorFlow models for efficiency
- Utilized SAS to perform statistical analysis on patient outcomes, identifying key factors that led to a 7% reduction in readmission rates.
- Implemented a dashboard system using Power BI to track and report on healthcare quality metrics, including patient satisfaction scores, readmission rates, and adherence to clinical guidelines..
- Implemented changes based on A/B test results, leading to a 15% improvement in patient outcomes and a 10% reduction in treatment costs. Employed Python (Pandas) for data manipulation and preprocessing, ensuring data quality and consistency.
- Integrated Electronic Health Record (EHR) data into analytical workflows to ensure a comprehensive view of patient health and medical history.

- Utilized Amazon Redshift for high-performance data warehousing, enabling rapid querying and analysis of large healthcare datasets for business intelligence and reporting purposes.

**HCL – India**

**Jan 2020 - Jul 2022**

### **Data Analyst**

- Regularly executed data cleaning, transformation and pre-processing techniques to maintain data integrity, ensuring accuracy and reliability for all analytical outputs.
- Conducted A/B testing and hypothesis-driven analysis to assess the impact of various supply chain strategies on performance.
- Improved data cleansing processes by 25% through the development of custom Python scripts, reducing data preparation time and enhancing the accuracy of predictive models.
- Worked on various data mapping initiatives, ensuring seamless integration of data from source systems into Teradata, enhancing data accessibility and usability for analysis.
- Managed and delivered 100% of data management projects, including business need analysis and high-level data modelling, resulting in improved data accuracy and alignment with business objectives.
- Implemented a cloud-based data storage and processing solution using Amazon Web Services (AWS), including Amazon S3 for data storage, Amazon Athena for querying, and AWS Glue for ETL processes, resulting in a 50% reduction in infrastructure costs and improved scalability.
- Engaged in extensive data management activities, conducting thorough data analysis and gap assessments to identify areas for improvement and optimize data utilization.
- Analysed, documented and presented user survey results, leading to an 18% enhancement in customer communication strategies, thereby improving user engagement and satisfaction metrics.
- Automated visual workflows in Alteryx to streamline data transformation processes, reducing data processing time by 40% and improving consistency across 25+ datasets.
- Built interactive dashboards in Tableau to track KPIs in supply chain operations, enabling real-time monitoring and proactive problem-solving.
- Built and managed complex SQL reports to audit over \$1M in debt, tracking financial performance across 500+ borrowers and lenders, ensuring 100% compliance and accurate financial reporting.
- Architected a robust data warehouse utilizing SQL Server and Excel to efficiently store, organize and analyse over 50GB of data, ensuring high performance and scalability for future data needs.
- Utilized Python and libraries like Matplotlib to execute advanced data extraction, cleaning, and visualization techniques, transforming complex datasets into clear, actionable insights for stakeholders.
- Utilized Alteryx for advanced data pre-processing by managing complex joins, calculations and transformations, ensuring 98% data accuracy for visualization in Tableau and Power BI, improving report generation efficiency by 30%.
- Constructed sophisticated data models that extract actionable insights from customer data, resulting in a 14% increase in successful sales initiatives and informed business strategies.
- Utilized Tableau to develop interactive dashboards and comprehensive reports that track business performance metrics (KPIs), enabling stakeholders to make data-driven decisions.

### **Education:**

**Master of Business Analytics and Project Management** - University of Connecticut, Stamford, CT, USA

**Bachelor of Technology in Biotechnology** - Koneru Lakshmaiah Education Foundation, Andhra Pradesh India.

### **Certification**

**Jira fundamentals** (From University of Connecticut)

**Base SAS certified** (From KL University Coursera)

**Clinical data management** (From KL University Coursera)

